# SafeMAP: Analysis-Guided Planning and Validation for Safe C-to-Rust Translation

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***Abstract*—Automatic C-to-Rust translators often preserve C pointer conventions and emit unsafe Rust. SafeMAP takes a different approach. It records each migration decision, from candidate classification through behavioral testing, in an auditable plan. Its analyzer collects pointer and dependency facts. A deterministic synthesizer emits Rust only when a safe rule matches exactly. Accepted code must compile without unsafe operations, expose no raw pointers in its public API, and match the original C on recorded randomized inputs. The C oracle runs under AddressSanitizer and UndefinedBehaviorSanitizer. Six helpers from a 32-function LLVM development corpus pass every check, including 1,000 differential cases per generated project. We then froze the implementation and ran it once on 146 functions from three independently maintained C libraries. Dependency grouping produced 142 units; 70 were classified candidate-safe and 17 produced Rust, but none compiled, so the held-out accepted rate is 0/142. Raw C2Rust produced no output in three project-level attempts because of toolchain failures. One of three one-shot Gemini requests produced retained Rust, which did not compile. The outcome-blind human label audit is still pending and its agreement and classification statistics are not reported. These results delimit the prototype's current generalization and the baseline comparison rather than demonstrating an effectiveness advantage.**

***Index Terms*—C-to-Rust migration, safe Rust, static analysis, program translation, software maintenance, memory safety**

## I. Introduction

C remains common in systems software. Its memory model leaves programmers responsible for preventing buffer overflows, use-after-free errors, null dereferences, and data races. Safe Rust checks ownership and borrowing at compile time while retaining low-level control. Translation is difficult because many C idioms have no direct safe-Rust equivalent. Raw pointers, output parameters, manual allocation, nullable values, and integer error codes require changes in representation and API design.

The established C2Rust translator preserves C behavior by generating Rust that often uses raw pointers and unsafe operations [1]. Its output is a useful reference, although it does not provide the guarantees of safe Rust. Prior work uses static analysis, refactoring, ownership-guided rewriting, transpiler redesign, and LLM-based rewriting to remove unsafe code after translation [2]–[8]. Other studies ask whether LLMs can translate directly to idiomatic Rust and repair failures with compiler or test feedback [9]–[13]. A third line targets safe Rust directly for structured subsets of C [14].

SafeMAP also targets safe Rust, but it is an experimental migration pipeline, not a general C compiler. Its acceptance rule is strict. Reducing unsafe code is insufficient, as is compilation with unsafe blocks or a C-style raw-pointer API. SafeMAP accepts a unit only when the generated Rust satisfies the project's safe-code policy and passes every applicable check. If no supported safe path exists, the tool rejects the unit and records why.

We ask:

- **RQ1:** For a restricted, statically identifiable subset of C idioms, how often can SafeMAP synthesize fully safe Rust?
- **RQ2:** Which C idioms are currently handled reliably by deterministic safe synthesis, and which idioms remain limiting?
- **RQ3:** How does SafeMAP compare with direct LLM safe translation and raw C2Rust under one acceptance policy?
- **RQ4:** Which individual planning and validation components contribute to accepted translation?
- **RQ5:** Which artifacts are required to audit and reproduce a safe-first migration decision?

The evaluation keeps development and test data separate. Development data contains 40 authored microbenchmarks, six authored regression modules, and ten LLVM programs whose results were already known. The frozen test corpus retains every function definition in preselected production source files from pinned revisions of inih, cJSON, and libcsv. The files remained sealed until the implementation was frozen.

This paper contributes:

- a C-to-Rust pipeline that accepts code only when it contains no unsafe Rust and exposes no raw pointers through its public API;
- a migration plan that records detected C idioms, proposed Rust APIs, eligibility decisions, and required checks;
- a deterministic synthesizer for supported C idioms, connected to compilation, source-policy checks, differential testing, and recorded sanitizer status; and
- a frozen function-level study with analyzer provenance, replayable test artifacts, component ablations, and direct

safe-output and C2Rust baselines; independent eligibility labels remain a separately reported pending audit.

## II. Background and related work

### A. C2Rust and safer-Rust refactoring

C2Rust translates C through a mostly syntactic process and is often used as a baseline or a migration starting point [1]. It preserves many C constructs with raw pointers and unsafe operations, so much of its output falls outside Rust's safe subset. Emre et al. found that raw pointers cause much of this unsafety [3]. Hong used static analysis to replace selected unsafe constructs in C2Rust output, including lock APIs and output parameters [4].

C2SaferRust combines C2Rust, static analysis, and LLM rewriting to reduce unsafe code in translated real-world programs [6]. Crown infers ownership models for C pointers and converts many of them to safe Rust types [5]. CRustS and Cp2SRust use different source-to-source transformations and transpiler designs [2], [7]. Industry studies and surveys report that automatic translation still needs substantial manual work and testing [8], [15]. SafeMAP uses C2Rust as a reference lane. Its primary lane generates final safe Rust directly for eligible units. Reductions in unsafe code remain diagnostic measurements; they count as migrations only when the final output passes the stated acceptance policy.

### B. LLM-based C-to-Rust translation

LLMs can produce idiomatic Rust, although correctness and type migration remain open problems. Hong and Ryu studied an LLM translator supplied with candidate signatures, caller and callee context, and compiler feedback [9]. C2RustTV uses LLMs for production and test-code translation, then checks the result with generated tests and repair loops [10]. Flourine evaluates LLM translation through cross-language differential fuzzing and finds lower success on more complex real-world samples [11]. RustFlow studies prompting and iterative repair [13]. Valenzuela et al. examine compiler errors in direct C-to-safe-Rust translation and in LLM repair of C2Rust output [12]. RustMap combines program analysis and LLM support at project scale [16].

### C. Safe-Rust translation for C subsets

Scylla translates an applicative C subset to safe Rust and shows how source structure can make safe generation possible [14]. SafeMAP likewise makes no claim that arbitrary C can be migrated automatically. It is a small research prototype built around migration plans, recorded rejections, and reproducible acceptance measurements.

### D. Validation and undefined behavior

Compilation alone cannot establish a safe migration. SafeMAP records compilation status, remaining unsafe constructs, raw pointers in public APIs, differential test results, C sanitizer results, and Miri status. Miri can detect undefined behavior during Rust execution [17]. ASan and UBSan check the executed C cases before SafeMAP uses them as behavioral oracles [18], [19]. Miri is disabled in the frozen experiment because the generated crates and installed toolchain do not provide a uniform applicable oracle; the artifact reports it as skipped rather than passed.

## III. SafeMAP design

### A. Outcome labels

SafeMAP records candidate classification and implementation support separately. The candidate label is `candidate_safe`, `manual_refactor_required`, `unsafe_required`, or `unknown`. A candidate-safe label means that screening found no known obstacle. It does not prove that a safe translation exists. The separate `implemented_support` field is true only when an exact synthesis rule matches. Later fields record generation, compilation, policy compliance, and behavioral test results. Every decision also identifies whether its facts came from libclang or the regular-expression fallback.

### B. Acceptance policy

SafeMAP reports policy and behavioral acceptance separately. A policy-safe unit must meet these conditions:

1) The final Rust code compiles with `#![forbid(unsafe_code)]`.
2) The final Rust contains no `unsafe` blocks, `unsafe fn`, `unsafe impl`, or `extern "C"` blocks.
3) The public API does not expose raw pointers such as `*const T` or `*mut T`.
4) The planned function or module unit is present in the final Rust output.

A behaviorally validated unit is policy-safe and passes an applicable C-versus-Rust oracle. A skipped or inapplicable check does not count as a pass. Cargo test and Clippy must pass when enabled. The frozen run records Miri as disabled and never treats that status as a pass. ASan and UBSan monitor the C oracle; a sanitizer finding disqualifies that execution as a correctness oracle. Partially migrated C2Rust output may still aid manual work, but SafeMAP does not accept it unless it passes the same policy.

### C. Pipeline

SafeMAP processes each C input in seven stages. Figure 1 shows both execution lanes and the policy they share:

1) Locate source files and recover compile information when possible.
2) Identify functions, signatures, pointer use, allocation patterns, unsupported constructs, and dependency structure.
3) Assign the screening taxonomy without consulting implemented synthesis rules.
4) Classify migration idioms such as output parameters, pointer-length arrays, nullable pointers, C strings, and return-code-plus-output patterns.
5) Build a plan that records the target Rust signature, ownership model, validation requirements, and reasons for any rejection.

6) Generate Rust when a deterministic rule supports the planned pattern.
7) Compile the result, check remaining unsafe code, count raw pointers, run differential tests, check ASan/UBSan on C, record Miri status, and export function-level measurements.

### D. Worked plan and rule selection

Consider the LLVM development helper:

```
double loop(float *x, float *y, long n) {
  double acc = 0.0;
  for (long i=0; i<n; ++i)
    acc += (double)x[i] * (double)y[i];
  return acc;
}
```

libclang finds two indexed pointers paired with `n`. It finds no indexed writes, calls, or dependencies. The classifier returns `candidate_safe`. The plan maps each pointer-length pair to a shared slice, and the `slice_dot_product` rule provides `implemented_support`. Since `loop` is a Rust keyword, the generated function uses a raw identifier:

```
pub fn r#loop(x: &[f32], y: &[f32]) -> f64 {
  x.iter().zip(y.iter())
   .map(|(&a,&b)| (a as f64)*(b as f64))
   .fold(0.0_f64, |s,v| s + v)
}
```

The initial value `+0.0` is necessary. Differential testing showed that a generic floating-point `sum()` could disagree with C on the sign bit of zero. The generated helper matches C bit for bit on 1,000 recorded cases and passes the Cargo, Clippy, ASan, and UBSan checks. LLVM's `doTest`, by comparison, calls allocation and memory APIs outside the supported subset. The classifier labels it `unsafe_required`, so synthesis does not run.

```
facts <- analyze(function)
candidate <- classify(facts)
unit <- SCC(call_graph, function)
plan <- safe_signature(facts) + idiom_facts
if candidate != candidate_safe:
    support <- not_applicable
else if exact_rule_matches(function, plan):
    support <- implemented_support
else: support <- not_implemented
generate only with implemented_support
report policy and behavior separately
```

### E. Supported idioms

The deterministic synthesizer currently handles: integer and floating scalar expressions, integer boolean and bit-reversal idioms, pointer-length and fixed-size arrays, scalar structs and field access, dependency-ordered internal helper calls, simple entry-point composition, mutable scalar pointer updates, output parameters, multiple output parameters, return-code plus output-parameter to `Result`, nullable pointers, reductions including two-slice floating dot products, simple C string lengths, simple allocation-to-`Vec` patterns, and single owned allocations to `Box`. It does not support unions, inline assembly, volatile memory access, unresolved aliasing, complex macros, custom allocators, function pointers, or pointer-integer casts.

### F. Baselines

SafeMAP compares its primary lane with two baselines. Raw C2Rust measures how much unmodified output passes the policy. The second baseline asks a fixed LLM to translate C directly to safe Rust without SafeMAP facts or plans. The frozen baseline is Google's `gemini-3.5-flash-lite`, accessed through its OpenAI-compatible API at temperature zero. The experiment keeps each prompt and response along with model settings, token counts, retries, and failures. It makes one request per project and performs no automatic retry; quota or provider failures remain in the denominator. Five ablations remove one component at a time: pointer-role classification, safe-signature generation, dependency grouping, idiom plans, or validation feedback. Every lane uses the same source units and behavioral checks.

## IV. Implementation

SafeMAP is a Python package. Separate modules handle ingestion, static analysis, migration planning, translation, repair, validation, and measurement. For a supported idiom, the deterministic synthesizer emits Rust under `#![forbid(unsafe_code)]`. The validation layer derives scalar and array harnesses from function signatures. It saves seeds and concrete C/Rust inputs and outputs, then records the result of each compiler, policy, differential, sanitizer, Clippy, and recorded Miri status. The benchmark runner writes CSV, Markdown, and LaTeX tables. One evaluation command combines SafeMAP, case-study, external-corpus, C2Rust, and ablation results. Its CSV files also record function roles, complete-project outcomes, outcome stages, and the status of each check.

The analyzer uses libclang when available. Otherwise it uses a pattern-based parser and records why the fallback was needed. Dependency analysis groups mutually dependent functions into one unit. Classification and planning finish before synthesis begins. The deterministic lane needs neither C2Rust nor a network model service.

## V. Evaluation methodology

### A. Datasets

The development corpus contains 40 authored microbenchmarks, six authored modules, and ten LLVM test-suite programs used during earlier experiments. The LLVM sources are pinned at commit `6cdc54e` [20]. They contain 589 lines of C and 32 functions. libclang analyzed every function. Their outcomes informed rule development, so they provide no held-out evidence.

The held-out test corpus uses pinned production sources from inih r62, cJSON v1.7.19, and libcsv commit `b1d5212`. A documented selection rule keeps every function in named production `.c` files without consulting SafeMAP output. Before execution, the experiment records repository URLs, revisions, source hashes, licenses, and physical line counts. A materialization script blocks access until the implementation has a frozen Git commit. This keeps test sources out of rule development.

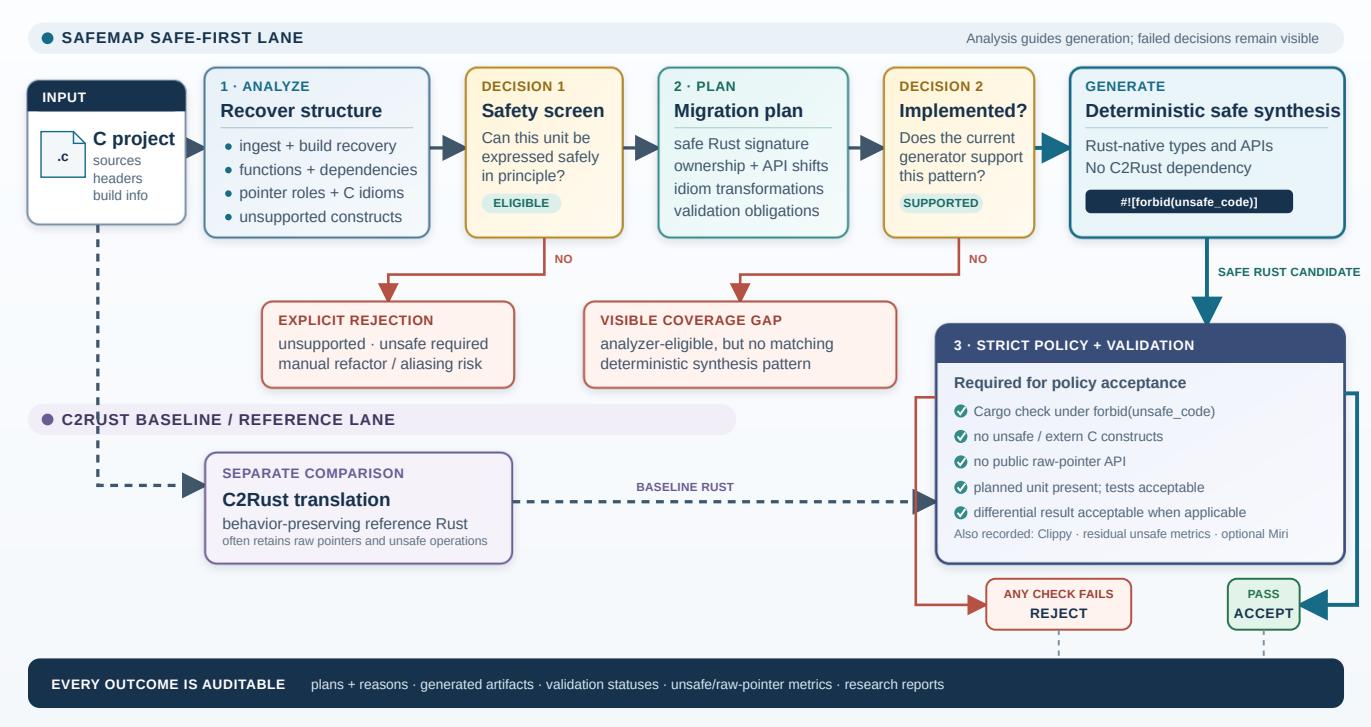



Fig. 1. SafeMAP conversion and acceptance pipeline. Static analysis records eligibility separately from synthesis support. The primary lane generates Rust from a migration plan; C2Rust runs in a separate reference lane. The same policy applies to both outputs. Reports retain unsupported cases, missing synthesis rules, and failed checks as distinct outcomes.

Two reviewers will label each held-out function without seeing the SafeMAP decision, then resolve disagreements. The labels use the candidate taxonomy. `unknown` cases are reported separately and excluded from binary precision and recall. The study reports raw agreement, Cohen's $\kappa$, disagreements by construct, and separate results for libclang and the fallback backend.

### B. Metrics

The primary effectiveness measure is the number of behaviorally validated, policy-safe units divided by all retained held-out units. The report also gives the count at each earlier stage: candidate-safe, supported, generated, compiling, and policy-safe. Results are separated by declared targets, helpers, entry points, other functions, and complete modules. Analyzer measurements include eligibility precision and error counts. Corpus tables report project and function counts, lines of code, cyclomatic complexity, pointer density, and construct frequencies.

### C. Execution snapshot

Before unsealing, the implementation and configuration were committed at `b2f455c`. A freeze record binds the experiment to that revision and records 153 passing pre-freeze tests. The held-out automatic evaluation then ran once, without post-outcome repair or retry. Artifacts record source and configuration hashes, tool versions, analyzer backend, commands, generated code, prompts and responses, random seeds, sanitizer status, timing, and failures. The direct-LLM baseline fixed its model and sampling settings before execution. The pre-freeze test count is not a held-out outcome.

TABLE I
DEVELOPMENT-ONLY OUTCOME CASCADE. "CAND." DENOTES `CANDIDATE_SAFE`; "BEHAV." DENOTES BEHAVIORALLY VALIDATED POLICY-SAFE OUTPUT.

| Corpus | Total | Cand. | Support | Gen. | Compile | Policy | Behav. |
|---|---|---|---|---|---|---|---|
| LLVM programs | 32 | 21 | 6 | 6 | 6 | 6 | 6 |
| Microbenchmarks | 81 | 76 | 36 | 36 | 36 | 36 | 36 |
| Authored modules | 25 | 25 | 19 | 19 | 19 | 19 | 17 |

## VI. Results

### A. Development evidence

The LLVM development corpus contains 32 functions from ten projects. After two analysis fixes and rule changes based only on development data, 21 functions receive `candidate_safe` and six receive `implemented_support`. The six generated functions compile, satisfy the policy, and pass behavioral testing. They are `sqr`, `fade`, `lerp`, the 32-bit and 64-bit reversal helpers, and the two-slice floating-point reduction. Before these changes, only `sqr` passed. The 6/32 result measures development progress and exercises the reporting pipeline. It cannot support a generalization claim because the same corpus guided the rule changes.

TABLE II
BEHAVIORALLY VALIDATED UNITS IN COMPONENT ABLATIONS. DENOMINATORS INCLUDE EVERY FUNCTION OR DEPENDENCY UNIT IN EACH DEVELOPMENT CORPUS.

| Configuration | Microbenchmarks | Modules |
|---|---|---|
| Full system | 36/81 | 17/25 |
| Without pointer roles | 16/81 | 3/25 |
| Without safe signatures | 6/81 | 0/25 |
| Without dependency grouping | 36/81 | 15/25 |
| Without idiom plans | 6/81 | 1/25 |
| Without validation feedback | 36/81 | 17/25 |

TABLE III
FROZEN HELD-OUT CORPUS CHARACTERIZATION.

| Project | C LOC | Functions | Parameters | Pointer params. | Complexity |
|---|---|---|---|---|---|
| cjson | 3191 | 113 | 202 | 167 | 632 |
| inih | 326 | 10 | 28 | 25 | 87 |
| libcsv | 545 | 23 | 52 | 37 | 132 |
| Total | 4062 | 146 | 282 | 229 | 851 |

TABLE IV
HELD-OUT SAFEMAP OUTCOME CASCADE. THE 146 FUNCTIONS FORM 142 DEPENDENCY-GROUPED TRANSLATION UNITS.

| Stage | Units |
|---|---|
| Retained translation units | 142 |
| Candidate-safe | 70 |
| Implemented support | 17 |
| Generated | 17 |
| Compiled | 0 |
| Policy-safe | 0 |
| Behaviorally validated | 0 |

TABLE V
HELD-OUT PROJECT-LEVEL BASELINE OUTCOMES UNDER THE SHARED POLICY.

| Lane | Attempts | Outputs | Compiled | Accepted |
|---|---|---|---|---|
| Raw C2Rust | 3 | 0 | 0 | 0 |
| Direct Gemini | 3 | 1 | 0 | 0 |

Two bugs show why the report keeps intermediate outcomes. The pointer screen first treated indexed reads as writes, which produced a false mutable-alias conflict for the floating-point reduction. It also parsed >> as a comparison and gave bit reversal a Boolean return type. Both bugs now have regression tests. Exact differential comparison found the separate signed-zero error described earlier.

On the 40 authored development programs, the full configuration validates 36 of 81 units, equal to 36 of 76 candidate-safe units, and 35 of 40 declared targets. Without pointer roles, the counts fall to 16/81 units and 15/40 targets. Without safe signatures or idiom plans, they fall to 6/81 and 5/40. Removing dependency grouping or validation feedback has no effect on this suite. These development ablations are diagnostic. The zero changes for the last two components show that the authored programs do not test their proposed benefits, so no causal conclusion follows.

A sixth authored module tests fixed arrays, a scalar struct, field access, two internal calls, five functions in one crate, and a small `main`. All five functions compile, pass policy checks, and pass the generated differential and sanitizer harness. Across six authored modules, the full system validates 17/25 units. Removing dependency grouping reduces the result to 15/25 because the two composed callers lose support. The module was written to test this mechanism, so the result is development evidence.

### B. Held-out corpus and primary outcome

The frozen corpus contains 4,062 physical lines of C and 146 functions. All functions were analyzed with libclang. Dependency grouping reduced them to 142 translation units; the no-grouping ablation retains all 146 separately.

SafeMAP classified 70/142 units as candidate-safe and matched an implemented synthesis rule for 17. All 17 produced Rust. The cJSON and libcsv project crates failed compilation because generated functions referred to untranslated project types or retained C expressions. inih had no supported generated unit. Consequently, no held-out unit reached policy or behavioral validation, and the primary accepted rate is 0/142. The C-oracle build also treated each library source as an executable and failed for lack of `main`; sanitizer and differential checks therefore could not rescue any unit. These are observed prototype and harness limitations, not excluded cases.

### C. Held-out baselines and ablations

Raw C2Rust was invoked once for each project and returned no translated output. Two logs identify failure to resolve standard C headers; the third records a failed invocation without a more specific reason. These failures remain in the denominator, but they prevent a meaningful output-quality comparison with SafeMAP. The direct Gemini baseline made three one-shot requests. Responses for cJSON and libcsv were rejected before validation for introducing, respectively, an `extern "C"` block and a raw pointer type. The retained inih response failed Cargo checking because it contained the C postfix increment operator. Raw provider logs record 32,790 input and 12,370 output tokens. No baseline output was accepted.

Removing pointer roles increased candidate classifications from 70 to 115 but reduced generated units from 17 to 10. Removing idiom plans reduced generation to one unit. Safe-signature removal and validation-feedback removal did not change the pre-compilation counts; disabling dependency grouping changed the denominator from 142 to 146 but not the 17 generated functions. Every configuration accepted zero units. The ablations therefore diagnose screening and synthesis dependencies, but provide no evidence that any component improves accepted held-out translation.

The two independent outcome-blind reviews have not yet been collected. Eligibility precision, recall, raw agreement, Cohen's $\kappa$, construct-level disagreements, and adjudicated

TABLE VI
HELD-OUT COMPONENT ABLATIONS. "SUPP." IS IMPLEMENTED SUPPORT.

| Configuration | Candidate | Supp. | Generated | Accepted |
|---|---|---|---|---|
| Full system | 70 | 17 | 17 | 0 |
| Without pointer roles | 115 | 10 | 10 | 0 |
| Without safe signatures | 70 | 17 | 17 | 0 |
| Without dependency grouping | 70 | 17 | 17 | 0 |
| Without idiom plans | 70 | 1 | 1 | 0 |
| Without validation feedback | 70 | 17 | 17 | 0 |

labels are therefore pending and must not be inferred from SafeMAP's own classifications.

## VII. DISCUSSION

### A. How the acceptance rule changes comparison

Code that compiles while exposing raw pointers through its public API has not reached safe Rust. The same policy therefore applies to every experimental lane. This prevents raw C2Rust output from being counted beside policy-safe output merely because both compile. C2Rust remains useful as a reference, and its output passes only when it meets the shared policy.

### B. Isolating planning components

The synthesizer requires a plan, so removing every form of guidance guarantees failure. The main ablation instead removes one input per run: pointer roles, safe signatures, dependency grouping, idiom plans, or validation feedback. A no-guidance run remains as a dependency check. Source units and validators stay fixed across runs.

### C. Recorded rejection

Unsupported constructs such as unions, volatile access, inline assembly, and function pointers occur in real C programs. Partial output can hide their effect on a migration. SafeMAP records each unsupported case as an outcome. The record shows where automatic processing stopped and where manual review must begin.

### D. Eligibility and synthesis coverage

The authored suites contain deliberately supported patterns, so they measure regression coverage. In the LLVM development set, 15 of 21 candidate-safe functions still lack a synthesis rule. In the held-out run, 53 of 70 candidate-safe units lack implemented support, and none of the 17 generated units compile. Candidate labels, synthesis coverage, and validation must therefore remain separate. Held-out human labels are still needed to estimate classification error.

## VIII. THREATS TO VALIDITY

**Benchmark representativeness.** The authored and LLVM corpora are development data. The held-out set improves project diversity but contains three small C libraries instead of large applications. We report size, complexity, pointer use, and construct frequencies so readers can judge the scope. The results do not cover arbitrary C.

**Implementation maturity.** SafeMAP is a research prototype. The analyzer and synthesizer use patterns and do not implement full C semantics. The tool reports unsupported constructs, and its supported subset remains small.

**Differential testing limits.** Differential testing over bounded random inputs is evidence, not equivalence proof. Generated harnesses cover supported scalar and array shapes. They do not cover every heap, callback, or stateful API. Saved seeds and outputs allow replay. ASan and UBSan reduce the chance of using an undefined C execution as an oracle, but they cannot say whether unexecuted inputs are defined.

**Label subjectivity.** Candidate safety sometimes requires judgment about whether an API redesign is automatic or manual. The planned outcome- blind reviews retain disagreements, adjudication, confidence scores, and $\kappa$, but those reviews are not yet complete. Until they are, the paper makes no precision, recall, or agreement claim.

**Toolchain sensitivity.** C2Rust, Clang, Rust, Clippy, and Miri can behave differently across host versions and configurations. The artifact records skipped and failed stages. Reproduction across machines will need tighter toolchain pinning. In this run C2Rust failed before emitting output, so its zero is an operational result rather than evidence about output safety.

**Harness completeness.** The production inputs are libraries, but the frozen C oracle attempted to link each source as an executable. The resulting missing-`main` failures made differential and sanitizer checks inapplicable. This limitation remains in the reported denominator because changing the harness after observing held-out outcomes would break the freeze.

**Metric interpretation.** Units are functions or dependency groups, not lines of code or whole-project migration percentages. Analyzer eligibility may exceed implemented synthesis support. Dataset size and complexity describe the corpus but do not remove this measurement limit.

## IX. FUTURE WORK

Current rules handle fixed arrays, scalar structs and fields, internal calls, small entry-point compositions, and multi-function crates. Planned work covers stateful APIs, callbacks, nested aggregates, unions with safe tagged representations, and aliases that require flow-sensitive borrowing analysis. Test generation also needs structured inputs, buffers with semantic length constraints, and preconditions derived from metadata. If held-out results lead to rule changes, the next generalization study will need a new frozen corpus.

## X. CONCLUSION

SafeMAP records each stage of C-to-Rust migration instead of reducing the result to compilation success. Classification, synthesis support, policy checks, and behavioral tests have separate outcomes. A missing test harness cannot count as a behavioral pass. Six functions in the LLVM development corpus pass every stage. In the frozen held-out run, 17 of 142 units generated Rust but none compiled or passed the

acceptance policy. C2Rust emitted no output under the frozen toolchain, and Gemini produced no compiling output. The ablations expose screening and synthesis dependencies but have identical zero acceptance. The study therefore finds that the prototype and frozen validation harness do not yet generalize to these production libraries. Independent human eligibility review remains necessary before reporting classification accuracy or reviewer agreement.

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